

HDOC Day-40

Student Name Operations Using Strings, Functions, and Call by Reference in C

Aim

To write a menu-driven C program that reads a student's first, middle, and last name and performs the required string operations using separate user-defined functions and pointers for call-by-reference results.

Header Files Used

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
```

Algorithm

1. Read the student's first, middle, and last name.
2. Combine the names using strcpy() and strcat().
3. Display the eight menu options and accept a choice.
4. Use strlen() to find the full-name length through a pointer.
5. Copy the full name using strcpy().
6. Compare first and last names using strcmp().
7. Construct the reverse order as last, middle, first.
8. Count vowels, consonants, digits, and spaces using ctype.h functions and pointers.
9. Search a word in the full name using strstr().
10. Repeat until Exit is selected.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void combineNames(const char f[], const char m[], const char l[], char full[]);
void findLength(const char full[], size_t *length);
void copyName(const char source[], char destination[]);
void compareNames(const char f[], const char l[], int *result);
void reverseNameOrder(const char f[], const char m[], const char l[], char rev[]);
void countCharacters(const char full[], int *v, int *c, int *d, int *s);
void searchWord(const char full[], const char word[], int *result);

int main()
{
    char first[50], middle[50], last[50];
    char full[160], copied[160], reverse[160], word[50];
    int choice, cmp, found, vowels, consonants, digits, spaces;
    size_t length;

    printf("Enter first name: "); scanf("%49s", first);
    printf("Enter middle name: "); scanf("%49s", middle);
    printf("Enter last name: "); scanf("%49s", last);

    combineNames(first, middle, last, full);

    do
    {
        printf("\n----- STUDENT NAME OPERATIONS -----\n");
        printf("1. Combine Names\n");
        printf("2. Find Length of Full Name\n");
        printf("3. Copy Full Name\n");
        printf("4. Compare First and Last Name\n");
        printf("5. Reverse Name Order\n");
```

```

printf("6. Count Vowels, Consonants, Digits and Spaces\n");
printf("7. Search a Word in Full Name\n");
printf("8. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        combineNames(first, middle, last, full);
        printf("Full Name: %s\n", full);
        break;
    case 2:
        findLength(full, &length);
        printf("Length of Full Name: %zu\n", length);
        break;
    case 3:
        copyName(full, copied);
        printf("Copied Name: %s\n", copied);
        break;
    case 4:
        compareNames(first, last, &cmp);
        if (cmp == 0)
            printf("First name and last name are equal.\n");
        else
            printf("First name and last name are different.\n");
        break;
    case 5:
        reverseNameOrder(first, middle, last, reverse);
        printf("Reverse Name Order: %s\n", reverse);
        break;
    case 6:
        countCharacters(full, &vowels, &consonants, &digits, &spaces);
        printf("Vowels: %d\n", vowels);
        printf("Consonants: %d\n", consonants);
        printf("Digits: %d\n", digits);
        printf("Spaces: %d\n", spaces);
        break;
    case 7:
        printf("Enter word to search: ");
        scanf("%49s", word);
        searchWord(full, word, &found);
        if (found) printf("Word found in the full name.\n");
        else printf("Word not found in the full name.\n");
        break;
    case 8:
        printf("Exiting program...\n");
        break;
    default:
        printf("Invalid choice! Please enter 1 to 8.\n");
}
} while (choice != 8);

return 0;
}

void combineNames(const char f[], const char m[], const char l[], char full[])
{
    strcpy(full, f); strcat(full, " "); strcat(full, m);
    strcat(full, " "); strcat(full, l);
}

void findLength(const char full[], size_t *length)
{
    *length = strlen(full);
}

void copyName(const char source[], char destination[])
{
    strcpy(destination, source);
}

void compareNames(const char f[], const char l[], int *result)
{
    *result = strcmp(f, l);
}

void reverseNameOrder(const char f[], const char m[], const char l[], char rev[])
{
    strcpy(rev, l); strcat(rev, " "); strcat(rev, m);
    strcat(rev, " "); strcat(rev, f);
}

void countCharacters(const char full[], int *v, int *c, int *d, int *s)
{
    int i;

```

```

char ch;
*v = *c = *d = *s = 0;

for (i = 0; full[i] != '\0'; i++)
{
    ch = full[i];
    if (isalpha((unsigned char)ch))
    {
        ch = tolower((unsigned char)ch);
        if (ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u') (*v)++;
        else (*c)++;
    }
    else if (isdigit((unsigned char)ch)) (*d)++;
    else if (isspace((unsigned char)ch)) (*s)++;
}
}

void searchWord(const char full[], const char word[], int *result)
{
    *result = (strstr(full, word) != NULL);
}

```

Test Case Table

Test	Operation	Example Input	Expected Result
1	Combine Names	Adeep Kumar Singh	Adeep Kumar Singh
2	Find Length	Adeep Kumar Singh	Length displayed
3	Copy Name	Adeep Kumar Singh	Copied name identical
4	Compare Names	Adeep / Singh	Names different
5	Reverse Order	Adeep Kumar Singh	Singh Kumar Adeep
6	Character Count	Adeep Kumar Singh	Counts displayed
7	Search Word	Kumar	Word found
8	Search Word	Hello	Word not found
9	Invalid Choice	9	Invalid choice message
10	Exit	8	Program exits

Sample Output

```

Enter first name: Adeep
Enter middle name: Kumar
Enter last name: Singh

Enter your choice: 1
Full Name: Adeep Kumar Singh

Enter your choice: 5
Reverse Name Order: Singh Kumar Adeep

Enter your choice: 7
Enter word to search: Kumar
Word found in the full name.

Enter your choice: 8
Exiting program...

```

Result

The menu-driven C program was successfully implemented using separate user-defined functions, strings, pointers, and call by reference. It combines names, finds length, copies and compares strings, reverses name order, counts vowels, consonants, digits and spaces, and searches for a word using the required string.h and ctype.h functions.